Sample Questions on Query Processing
Fall 2025

The student relation schema is given as follows:
Student (id, name, street, city, NID, year-admit, tot_cred, cgpa)

Q1: The size of the given NSU student relation is 2000 blocks. The seek time is 10ms and block transfer time is 0.1 ms. The query is

SELECT * FROM student WHERE id >=1801111042

The student relation is physically sorted on id. The number of blocks with id >= 1801111042 is 500.
The height of the primary index on id is 3.

Find the estimated cost for the query using the primary index.

Q2: Find the cost of query SELECT * FROM student WHERE id < 1801111042 using data of Q1.

Q3: Using student relation of Q1,
The SQL is SELECT * FROM student WHERE City='Dhaka'
There are B+ tree index on city and the height of the index is 3. The number of tuples (pointers) for City='Dhaka' is 200. Considering each tuple in separate block, find the least cost of the query using appropriate algorithms.

Q4:

The SQL is SELECT * FROM student WHERE CGPA=3 AND City='Dhaka' AND year-admit=2020
There are B+ tree indices on CGPA, city and year-admit.The heights of the indices are 3, 3 and 2 respectively. The number of tuples (pointers) for CGPA=3, City='Dhaka' and year-admit=2020 are 100, 200 and 1000 respectively. Consider each tuple in separate block.

a. Which algorithms will be applicable to process the above query?
b. Find the costs of each algorithm and mention the algorithm that will be used by query engine to process the query.

Q5:

The relational schema for student course registration are as follows

Student (<u>Sid</u>, name, street, city. Mobile, email, CGPA, age)
Takes (<u>course-id, Sid, semester, year</u>, GP)
Course (<u>course-id</u>, title, credit-hour)

The primary indices are given on all of the primary keys.

Five secondary indices are given on attributes name, street, city, age and title.

Explain how the following SQL will be executed using index?

i. SELECT *
FROM student
WHERE sid > 10001

ii. SELECT *
FROM course
WHERE title = 'Database'

iii. SELECT *
FROM takes
WHERE sid = 10005 and course_id = 'CSE311' and semester = 'Spring' and year = 2022

Q6:

The student relation is physically sorted on NID. Two indices: id and NID are given.

Explain the query using index and find the cost for SQL: SELECT * FROM student WHERE id = 1801111042. The height of id index is 3.

Q7: The student and takes schema are: 4

Student (id, name, CGPA, street, city, year-admit)

Takes (id, course-id, semester, year)

The number of records and sizes of student and takes relation are as follows:

Relation	**No. of records**	**No. of blocks**
student	1000	50
takes	5000	10

a. Show the algorithm for student ⋈ takes using nested loop join.

b. Show the algorithm for student ⋈ takes using block nested loop join.
c. Derive the worst case and best-case cost of natural join operation student ⋈ takes using nested loop join.
d. Derive the worst case and best-case cost of natural join operation student ⋈ takes using block nested loop join.

Q9: Write relational algebra and construct the execution tree for the following:

The relational schema for student course registration are as follows

Student (Sid, name, street, city. Mobile, email, CGPA, age)
Takes (course-id, Sid, semester, year, GP)
Course (course-id, title, credit-hour)

(a)

SELECT sid, name

FROM student inner join takes ON student.sid = takes.sid

WHERE city = ‘Dhaka’ AND year = 2023

(b)

SELECT sid, name

FROM (student inner join takes ON student.sid = takes.sid) inner join course ON takes.course_id = course.course_id

WHERE city = ‘Dhaka’ AND year = 2023 AND credit-hour = 3

Q10: Explain the above queries processing using i) materialized method ii) Pipeline method